

Jiayi Zhang

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Education

University of Nottingham Ningbo China (UNNC)

BSc (HONS) MATHEMATICS WITH APPLIED MATHEMATICS

- GPA - 3.94/4.00

Ningbo China

Sep 2024 - Jun 2028 (Expected)

Research Interests

My research focuses on **foundation-model-based robot learning** for reliable, **long-horizon manipulation**. I study how foundation models, structured task-and-motion reasoning, and **geometry-aware visuomotor policies** can be integrated through effective **action representations**, reusable skill abstractions, and policy-controller co-design. My goal is to enable robots to ground semantic goals into closed-loop physical behaviors, adapt and recover under uncertainty, and generalize to complex human-centered tasks such as **tabletop Mahjong**.

Experience

Tsinghua University (THU)

RESEARCH ASSIST, ADVISED BY [PROF. MENGDI XU](#)

- Research Topics:** Manipulation, In-Context Learning, Robot learning

Beijing, China

July 2026- Now

Shanghai Jiao Tong University (SJTU)

RESEARCH ASSIST AT [IWIN-FINS LAB](#), ADVISED BY [PROF. JIANPING HE](#)

- Research Topics:** Robot learning, Control, Manipulation.

Robot arm: Project leader

D.I.V.E FORUM: Organizer

Underwater 3D reconstruction: Project leader

WBCD 2nd dual arm robot: member

Shanghai, China

June 2025 - June 2026

The University of Hong Kong (HKU)

RESEARCH INTERN AT [IoT LAB](#), ADVISED BY [PROF. EDITH C.H. NGAI](#)

- Research Topics:** federated learning, fairness, spatio-temporal graphs.

Hong Kong SAR, China

March 2025 – August 2025

Publications

PREPRINTS

[P1] F²STNet: Fair and Federated Spectral-Temporal Modeling for Graph Forecasting

Jiayi Zhang*, Jinfeng Xu*, Hewei Wang*, Siyuan Cen, Haidong Huang, Yiyao Zhan, Zheyu Chen, Jinjiang You, Ai Jian, Edith C. H. Ngai
Under review, 2025

CONFERENCE PROCEEDINGS

[C4] Multi-modal Dynamic Proxy Learning for Personalized Multiple Clustering

Jinfeng Xu, Zheyu Chen, Shuo Yang, Jinze Li, Ziyue Peng, Zewei Liu, Hewei Wang, [Jiayi Zhang](#), Edith C. H. Ngai
AAAI, 2026

[C3] Multi-Cali Anything: Dense Feature Multi-Frame Structure-from-Motion for Large-Scale Camera Array Calibration

Jinjiang You*, Hewei Wang*, Yijie Li, Mingxiao Huo, LVT Ha, Mingyuan Ma, Jinfeng Xu, [Jiayi Zhang](#), Puzhen Wu, Shubham Garg, Wei Pu
IROS, 2025 (Oral) [\[Paper\]](#)

[C2] Spec-LLAVA: Accelerating Vision-Language Models with Dynamic Tree-Based Speculative Decoding

Mingxiao Huo*, [Jiayi Zhang*](#), Hewei Wang*, Jinfeng Xu, Zheyu Chen, Huilin Tai, Ian Yijun Chen
ICML Workshop, 2025 [\[Paper\]](#)

[C1] Segregation and Context Aggregation Network for Real-Time Cloud Segmentation

Yijie Li*, Hewei Wang*, [Jiayi Zhang](#), Jinjiang You, Jinfeng Xu, Puzhen Wu, Yunzhong Xiao, Soumyabrata Dev
ICLR Workshop, 2025 [\[Paper\]](#)

JOURNAL ARTICLES

[J2] TrafficKAN-GCN: Graph Convolutional-based Kolmogorov-Arnold Network for Traffic Flow Optimization

[Jiayi Zhang*](#), Yiming Zhang*, Yuan Zheng, Yuchen Wang, Jinjiang You, Yuchen Xu, Wenxing Jiang, Soumyabrata Dev
arXiv preprint, 2025 [\[Paper\]](#)

Honors & Awards

2026	Outstanding student (Top 1% student) , UNNC Award for Outstanding Students (UG)	Ningbo China
2025	Provost's Scholarship (Top 5% student) , University Academic Excellence Scholarship	Ningbo China
2025	Third Prize & Best Researcher Award , 9th Golden Interns Award	Ningbo China
2025	Honorable Mention , Interdisciplinary Contest in Modeling 2025	Online

Academic Services

Reviewer NeurIPS, AAAI, AISTATS 2025–Present

Leadership/Mentorship/Social

Semester Peer Mentor

UNIVERSITY OF NOTTINGHAM NINGBO CHINA (UNNC)

- Provided academic guidance and practical resources for freshmen.

Ningbo China
Sep. 2025–Jun. 2026

Skills

Programming Python, C++, $\LaTeX$, MATLAB, HTML, CSS, R

Frameworks PyTorch, Tensorflow, NumPy, Flask, MySQL, Git, Anaconda, OpenCV, ROS

Models CNN, ResNet, Faster R-CNN, YOLO, LSTM, Transformer, BERT, ViT, GANs, VAE, Stable Diffusion

Platforms Unitree Z1 Pro, Franka Research 3, AgileX Piper

Additional

Language Chinese (native), English (working proficiency)

Interests Running, hiking, mahjong, chessing, pickleball, roguelike and strategy games